

Influence of tea drinking on manganese intake, manganese status and leucocyte expression of MnSOD and cytosolic aminopeptidase P.



OBJECTIVE: Since black tea contains high levels of manganese (Mn), we investigated the relationship between dietary Mn intake, circulating Mn levels and leucocyte expression of two Mn-dependent enzymes in tea drinkers and non-tea drinkers. DESIGN: We assessed Mn intakes (food frequency questionnaire), fasting whole blood and plasma Mn levels, and quantitative expression of peripheral blood mononuclear cell Mn-dependent superoxide dismutase (MnSOD) and cytosolic aminopeptidase-P (cAP-P). SETTING AND SUBJECTS: In total, 24 tea drinkers (≥ 1 l black tea/day) and 28 non-tea drinkers were recruited from the staff and students of King's College London by circular email. RESULTS: Dietary Mn intakes (mean (range)) were significantly lower ($P < 0.0001$) in non tea drinkers (3.2 mg/day (0.5-6.5)) than tea drinkers (5.5 mg/day (2-12) or 10 mg/day (5-20) depending upon the value used for Mn levels of black tea). Whole blood, plasma Mn levels and expression of MnSOD and cAP-P did not differ between the groups. In a continuous analysis, whole blood Mn levels and expression of MnSOD correlated inversely but no other parameters associated with each other. CONCLUSIONS: Tea drinking is a major source of dietary Mn and intakes commonly exceed proposed adequate intake values of 1.8-2.3 mg Mn/day and, on occasion, exceed upper limits of 10-11 mg/day. Dietary Mn intake has little influence on markers of Mn status or expression of Mn-dependent enzymes. Fasting whole blood Mn levels and leucocyte expression of MnSOD could, together, be further investigated as markers of Mn status.